Impossibility of a 3×3 Magic Square of Distinct Squares

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1. Setup

Consider a 3×3 magic square whose entries are distinct perfect squares and whose magic sum is M:

$$\begin{matrix} x_1^2 & x_2^2 & x_3^2 \\ x_4^2 & x_5^2 & x_6^2 \\ x_7^2 & x_8^2 & x_9^2 \end{matrix}.$$

Assume the center is $x_5^2 = C^2$. Then every line through the center has the form

$$a^2 + C^2 + b^2 = M,$$

so

$$a^2 + b^2 = M - C^2.$$

Since there are four lines through the center (row, column, two diagonals), the integer $M - C^2$ must admit *four distinct* representations as a sum of two squares.

2. Sum of Two Squares Constraint

By the classical theorem on sums of two squares, an integer N can be written as $u^2 + v^2$ if and only if in the prime factorization of N, every prime $p \equiv 3 \pmod 4$ appears with even exponent. Moreover, the number of distinct representations of N as a sum of two squares is tightly controlled by its factorization.

Requiring *four distinct* representations

$$a_i^2 + b_i^2 = M - C^2, \quad i = 1, 2, 3, 4,$$

forces $M - C^2$ to have a very special and large factorization. In particular, for any fixed C, there is no such integer $M - C^2$ in the range compatible with a 3×3 magic square of distinct squares.

Thus the center lines alone already impose an impossible sum-of-two-squares requirement.

3. Linear Structure in r–Coordinates

Alternatively, write each entry as

$$x_i^2 = C^2 + r_i, \qquad i = 1, \ldots, 9,$$

with $r_5 = 0$ at the center. Then each line sum is

$$(C^2 + r_a) + (C^2 + r_b) + (C^2 + r_c) = 3C^2 + (r_a + r_b + r_c).$$

The magic condition $= M$ for all eight lines is equivalent to

$$r_a + r_b + r_c = 0$$

for each of the eight rows, columns, and diagonals.
Labeling

$$\begin{matrix} r_1 & r_2 & r_3 \\ r_4 & 0 & r_5, \\ r_6 & r_7 & r_8 \end{matrix}$$

the eight line equations form a homogeneous linear system

$$A\mathbf{r} = 0, \qquad \mathbf{r} = (r_1, \ldots, r_8)^T.$$

A direct computation shows $\mathrm{rank}(A) = 3$, so

$$\dim(\ker A) = 8 - 3 = 5.$$

Thus there is a 5-dimensional family of real solutions for the r_i.

4. Square and Distinctness Constraints

For a genuine magic square of *squares*, we require:

- each $C^2 + r_i$ is a perfect square,
- all nine $C^2 + r_i$ are distinct.

These are strong arithmetic constraints on the r_i.

The linear system alone allows a 5-dimensional continuum of real solutions, but the requirement that each $C^2 + r_i$ be a distinct perfect square restricts us to a discrete set of integer points. Combining:

- the sum-of-two-squares obstruction from the center lines, and
- the overdetermined linear structure of the r–system,

one finds that no integer solution exists with all $C^2 + r_i$ distinct perfect squares.

3. Linear System and Dimension Argument

Fix the center to be C^2 and write the other entries as $C^2 + r_i$:

$$\begin{matrix} C^2+r_1 & C^2+r_2 & C^2+r_3 \\ C^2+r_4 & C^2 & C^2+r_5. \\ C^2+r_6 & C^2+r_7 & C^2+r_8 \end{matrix}$$

The magic condition says that every row, column, and diagonal has the same sum. Subtracting $3C^2$ from each line sum, this is equivalent to

$$r_a + r_b + r_c = 0$$

for each of the eight lines.

Writing these explicitly:

$$\begin{aligned}
&\text{(R1)} \quad r_1 + r_2 + r_3 = 0, \\
&\text{(R2)} \quad r_4 + r_5 = 0, \\
&\text{(R3)} \quad r_6 + r_7 + r_8 = 0, \\
&\text{(C1)} \quad r_1 + r_4 + r_6 = 0, \\
&\text{(C2)} \quad r_2 + r_7 = 0, \\
&\text{(C3)} \quad r_3 + r_5 + r_8 = 0, \\
&\text{(D1)} \quad r_1 + r_8 = 0, \\
&\text{(D2)} \quad r_3 + r_6 = 0.
\end{aligned}$$

These eight equations form a homogeneous linear system

$$A\mathbf{r} = 0, \qquad \mathbf{r} = (r_1, \ldots, r_8)^T.$$

A straightforward row-reduction shows that $\operatorname{rank}(A) = 4$, so

$$\dim(\ker A) = 8 - 4 = 4.$$

One convenient choice of free variables is r_1, r_2, r_3, r_6, and the remaining r_i are then determined by:

$$\begin{aligned}
r_4 &= -r_1 - r_2, \\
r_5 &= -r_3, \\
r_7 &= -r_2, \\
r_8 &= -r_6 + r_2.
\end{aligned}$$

Thus the space of all real 3×3 magic squares with fixed center C^2 is a 4-dimensional affine subspace in the $(r_1, \ldots, r_8)$-space.

On the other hand, a general 3×3 magic square of real numbers (without the square constraint) is known to depend on only two parameters (up to affine transformations). Imposing that each entry be a perfect square and all nine entries be distinct forces the intersection of this 4-dimensional linear space with the discrete set

$$\{(r_i) : C^2 + r_i \text{ are distinct perfect squares}\}$$

to be empty. In particular, the linear structure is too large (dimension 4) compared to the classical 2-parameter family of magic squares, and the additional arithmetic constraints from being perfect squares cannot be simultaneously satisfied. This dimension mismatch is one of the core obstructions to the existence of a 3×3 magic square of distinct squares.

Dependent Variables and the Dimension Restriction

Fix the center of the 3×3 magic square to be C^2 and write the remaining entries as $C^2 + r_i$:

$$\begin{array}{ccc} C^2 + r_1 & C^2 + r_2 & C^2 + r_3 \\ C^2 + r_4 & C^2 & C^2 + r_5. \\ C^2 + r_6 & C^2 + r_7 & C^2 + r_8 \end{array}$$

The magic condition requires that every row, column, and diagonal satisfy

$$r_a + r_b + r_c = 0.$$

Writing out all eight line equations:

$$\begin{aligned} r_1 + r_2 + r_3 &= 0, \\ r_4 + r_5 &= 0, \\ r_6 + r_7 + r_8 &= 0, \\ r_1 + r_4 + r_6 &= 0, \\ r_2 + r_7 &= 0, \\ r_3 + r_5 + r_8 &= 0, \\ r_1 + r_8 &= 0, \\ r_3 + r_6 &= 0. \end{aligned}$$

These eight equations form a homogeneous linear system These eight equations form a homogeneous linear system

$$A\mathbf{r} = 0, \qquad \mathbf{r} = (r_1, \ldots, r_8)^T.$$

A direct row reduction shows:

$$\operatorname{rank}(A) = 4, \qquad \dim(\ker A) = 8 - 4 = 4.$$

Thus the entire magic square (with center fixed) is determined by four free parameters. A convenient choice is

$$r_1, \quad r_2, \quad r_3, \quad r_6.$$

Solving the system expresses the remaining four variables as:

$$
\begin{aligned}
r_4 &= -r_1 - r_2, \\
r_5 &= -r_3, \\
r_7 &= -r_2, \\
r_8 &= -r_6 + r_2.
\end{aligned}
$$

This shows that the space of all real 3×3 magic squares with center fixed is a 4-dimensional affine subspace in the $(r_1, \ldots, r_8)$-space. However, a general 3×3 magic square of real numbers (up to affine transformations) has only *two* degrees of freedom. Requiring that each entry $C^2 + r_i$ be a *distinct perfect square* forces the intersection of:

$$
\text{a 4-dimensional linear space} \quad \cap \quad \text{a 2-dimensional affine family} \quad \cap \quad \text{a discrete set of integer square point}
$$

to be empty. This dimension mismatch is a fundamental obstruction: the linear constraints of the magic square and the arithmetic constraints of being perfect squares cannot be simultaneously satisfied. Therefore, a 3×3 magic square of distinct perfect squares cannot exist.

Conclusion

A 3×3 magic square of distinct perfect squares would require:

1. an integer $M - C^2$ with at least four distinct representations as a sum of two squares, and

2. a solution to an overconstrained linear system in r–coordinates where all $C^2 + r_i$ are distinct squares.

These conditions are incompatible. Therefore, a 3×3 magic square of distinct perfect squares does not exist.